

Multiplicative Inverse, Fermat's little Theorem

Michael L. Littman

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Overview

Arithmetic with a Prime Modulus (8.6)

 Multiplicative Inverses (8.6.1)

 Cancellation (8.6.2)

 Fermat's Little Theorem (8.6.3)

Back to basics

Definition: The *multiplicative inverse* of a number x is a number x^{-1} such that: $x \cdot x^{-1} = 1$.

Division by x is really multiplication by x^{-1} .

Useful for solving “mod n ” equations. That's because

$a \cdot x \equiv b \pmod{n}$ iff

$a^{-1} \cdot a \cdot x \equiv a^{-1} \cdot b \pmod{n}$ iff

$1 \cdot x \equiv a^{-1} \cdot b \pmod{n}$

General rule for existence of multiplicative inverses? Perhaps a has an inverse iff $\gcd(a, n) = 1$?

Inverse mod prime

If this rule holds, all values (except zero!) have inverses mod a prime.

Lemma: If p is prime and k is not a multiple of p , then k has a multiplicative inverse modulo p .

Proof: Since p is prime and k is not a multiple of p , $\gcd(p, k) = 1$. Therefore, there are s and t such that $1 = sp + tk$. So, mod p , that's $1 = tk$, or $t = k^{-1} \bmod p$. QED.

Example: What's the multiplicative inverse of 3 (mod 11)?

$$\text{gcdcombo}(3, 11) = (4, -1, 1)$$

So? 4 works. Because $1 = 4 \times 3 - 1 \cdot 11$, so, mod 11, that's $1 = 4 \times 3$.

Back to dividing both sides

Earlier, we saw:

$$7 \equiv 28 \pmod{3}$$

$$1 \equiv 4 \pmod{3} \quad \text{divide by 7}$$

Doesn't actually work, in general:

$$12 \equiv 6 \pmod{3}$$

$$4 \not\equiv 2 \pmod{3} \quad \text{divide by 3}$$

Why? Because we're really talking about multiplying both sides by 0^{-1} , which doesn't exist.

Apart from dividing by 0, we can cancel.

Cancellation proof

If we have

$$ak \equiv bk \pmod{p}$$

and p is prime and $k \not\equiv 0 \pmod{p}$, then $k^{-1} \pmod{p}$ exists.

Multiply both sides by k^{-1} and congruence is maintained.

Never need to multiply big numbers

When doing multiplication mod n , we can always mod n the numbers first.

Example:

$$\begin{aligned} &7415 \times 2993 \bmod 3 \\ &= 22193095 \bmod 3 \\ &= 1 \end{aligned}$$

OR:

$$\begin{aligned} &(7415 \bmod 3) \times (2993 \bmod 3) \bmod 3 \\ &(2 \times 2) \bmod 3 \\ &= 1. \end{aligned}$$

Proof

$$ab \bmod n = (a \bmod n)(b \bmod n) \bmod n.$$

$$a = q_1 n + r_1$$

$$b = q_2 n + r_2$$

$$ab = (q_1 n + r_1)(q_2 n + r_2)$$

$$ab = (q_1 q_2 n + q_1 r_2 + q_2 r_1)n + r_1 r_2$$

Solving an old equation

In an early lecture, we wanted to know if there's an n such that $n^2 \equiv 8 \pmod{10}$.

We don't quite have the ability to take square roots to solve this equation. But, we now know that if it's not true of n from 0 to 9, it's not true for any n . Why? Because you can take mod *before* multiplying.

Note also: Connor Jordan proved that the sequence of quadratic residuals (mods of squares n) will always be a length n palindrome!

$$x^2 \equiv (n - x)^2 \pmod{n}$$

$$\text{iff } x^2 \equiv n^2 - 2nx + x^2 \pmod{n}$$

$$\text{iff } x^2 \equiv x^2 \pmod{n}.$$

Permuting

Corollary: Suppose p is prime and k is not a multiple of p . Then, the sequence of remainders on division by p of the sequence:

$$1 \cdot k, 2 \cdot k, \dots, (p-1) \cdot k$$

is a permutation of the sequence:

$$1, 2, \dots, (p-1).$$

Example, $k = 3$, $p = 11$:

i	1	2	3	4	5	6	7	8	9	10
$\times k$	3	6	9	12	15	18	21	24	27	30
$\text{mod } p$	3	6	9	1	4	7	10	2	5	8

Permutation proof

Proof: The sequence of remainders contains $p - 1$ numbers. Since $i \times k$ is not divisible by p (neither contains a factor of p) for $i = 1, \dots, p - 1$, all these remainders are in $[1, p)$ by the definition of remainder. Furthermore, the remainders are all different. That's because no two numbers in $[1, p)$ are congruent modulo p . By the Cancellation property, $i \cdot k \equiv j \cdot k \pmod{p}$ iff $i \equiv j \pmod{p}$. Thus, the sequence of remainders must be a permutation of the numbers from 1 to $p - 1$. QED.

It's a magic shuffle function. Useful for randomization and sending secret messages!

Fermat's little theorem

Theorem: Suppose p is prime and k is not a multiple of p . Then:

$$k^{p-1} \equiv 1 \pmod{p}.$$

Proof:

$$(p-1)!$$

$$= 1 \cdot 2 \cdot \dots \cdot (p-1)$$

Defn. of factorial

$$= \text{rem}(k, p) \cdot \text{rem}(2k, p) \cdots \text{rem}((p-1)k, p)$$

Permutation lemma

$$\equiv k \cdot 2k \cdots (p-1)k \pmod{p}$$

Congruence of mult.

$$\equiv (p-1)!k^{p-1} \pmod{p}$$

algebra

Note that $(p-1)!$ is not a multiple of p because none of $1, 2, \dots, (p-1)$ contain a factor of p . So, by the Cancellation lemma, we can cancel $(p-1)!$ from the top and bottom, proving the claim. QED

Inverses from Fermat's little theorem

Since $k^{p-1} \equiv 1 \pmod{p}$ and $k^{p-1} = k \cdot k^{p-2}$, that tells us that k^{p-2} is the multiplicative inverse for k .

We can compute $k^{p-2} \pmod{p}$ efficiently using a technique called exponentiation by repeated squaring.

Running time is $2 \log p$, just like “gcdcombo”.

Exponentiation by Repeated Squaring Idea

Can always compute a^k by $k - 1$ multiplications of a .

If k is even, can compute it with $k/2 - 1$ multiplications of a to get $a^{k/2}$. Then, $a^k = (a^{k/2})^2$. So, one more multiplication and we're there.

If k is odd, similar trick to get $a^{(k-1)/2}$, then square, then multiply one more a .

Repeating this idea, the number of multiplications is on the order of $2 \log k$.

Exponentiation by Repeated Squaring

```
def repsq(a,k):  
    if k == 0: return(1)  
    if k % 2 == 0:  
        sqroot = repsq(a,k/2)  
        return(sqroot*sqroot)  
    sqrootdiva = repsq(a,(k-1)/2)  
    return(sqrootdiva*sqrootdiva*a)
```

Exponentiation by Repeated Squaring Mod Style

```
def repsqmodn(a,k,n):  
    a := a % n  
    if k == 0: return(1)  
    if k % 2 == 0:  
        sqrootdiva = repsqmodn(a,k/2,n)  
        return((sqrootdiva*sqrootdiva) % n)  
    sqrootdiva = repsqmodn(a,(k-1)/2,n)  
    return((sqrootdiva*sqrootdiva*a) % n)
```